

YOLOv11 Environment Setup

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[!IMPORTANT]

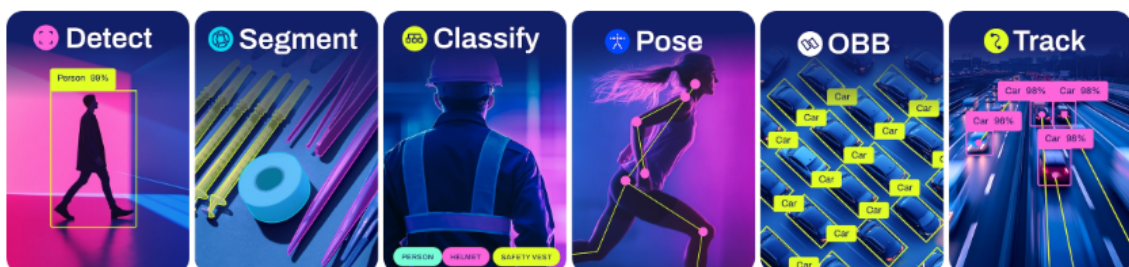
- The factory image has pre-configured the YOLOv11 environment. This tutorial is provided for users who need to set up the environment themselves. If you want to experience the complete functionality directly, you can skip this section!!!
- This tutorial is only for Jetson Orin and NX controllers, not applicable to Raspberry Pi or RDK X5.

0. YOLOv11 Introduction

1. YOLO (You Only Look Once) is a popular object detection and image segmentation model developed by Joseph Redmon and Ali Farhadi at the University of Washington. YOLO was launched in 2015 and has become widely popular for its high speed and accuracy.
2. Ultralytics' latest YOLO models deliver state-of-the-art (SOTA) performance across multiple tasks, including object detection, segmentation, pose estimation, tracking, and classification, and can be deployed in various AI applications and domains.
3. YOLOv11 is the latest version of the widely acclaimed real-time object detection and image segmentation model. Based on cutting-edge technologies in deep learning and computer vision, YOLO11 offers unparalleled performance in speed and accuracy. Its streamlined design makes it suitable for various applications and easily adaptable to different hardware platforms, from edge devices to cloud APIs.

🌟 Models

Ultralytics supports a wide range of YOLO models, from early versions like [YOLOv3](#) to the latest [YOLO11](#). The tables below showcase YOLO11 models pretrained on the [COCO](#) dataset for [Detection](#), [Segmentation](#), and [Pose Estimation](#). Additionally, [Classification](#) models pretrained on the [ImageNet](#) dataset are available. [Tracking](#) mode is compatible with all Detection, Segmentation, and Pose models. All [Models](#) are automatically downloaded from the latest Ultralytics [release](#) upon first use.





YOLOv11 documentation link: <https://docs.ultralytics.com>

1. Course Content

1. Master the setup method for YOLOv11 runtime environment

2. Prerequisites

```
sudo apt update
sudo apt install python3-pip -y
sudo pip install -U pip
```

3. Install Ultralytics

```
pip install -U ultralytics
```

4. Configure Torch Environment

The factory image has downloaded related torch function installation packages. If you are using your own environment, you need to search and download the corresponding version from the PyTorch or NVIDIA official website pages. The process of setting up a DIY environment will not be explained here.

PyTorch official website: <https://pytorch.org/get-started/locally/>

View torch installation packages

```
cd ~/software/torch_pkgs
ls
```

```
jetson@yahboom:~/software/torch_pkgs$ ls
tensorflow-2.16.1+nv24.08-cp310-cp310-linux_aarch64.whl  torchvision-0.20.0a0+afc54f7-cp310-cp310-linux_aarch64.whl
torch-2.5.0a0+872d972e41.nv24.08-cp310-cp310-linux_aarch64.whl
jetson@yahboom:~/software/torch_pkgs$
```

torch

```
pip install ./torch-2.5.0a0+872d972e41.nv24.08-cp310-cp310-linux_aarch64.whl
```

torchvision

```
pip install ./torchvision-0.20.0a0+afc54f7-cp310-cp310-linux_aarch64.whl
```

tensorflow

```
pip install ./tensorflow-2.16.1+nv24.08-cp310-cp310-linux_aarch64.whl
```

5. Verify Installation

Verify Ultralytics

```
python3 -c "import ultralytics; print(ultralytics.__version__)"
```

Verify Torch

```
python3 -c "import torch; print(torch.__version__);  
print(torch.cuda.is_available())"
```

Verify Torchvision

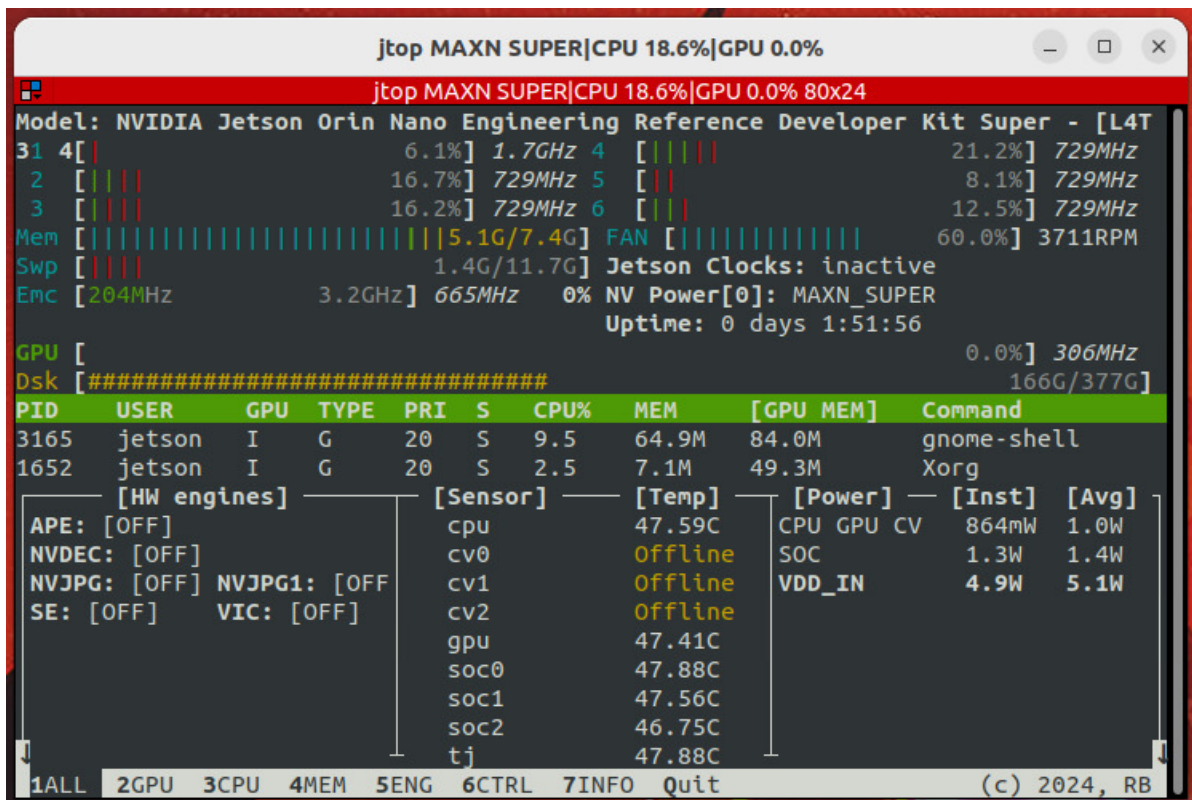
```
python3 -c "import torchvision; print(torchvision.__version__)"
```

```
jetson@yahboom:~/software/torch_pkgs$ python3 -c "import ultralytics; print(ultralytics.__version__)"  
8.3.214  
jetson@yahboom:~/software/torch_pkgs$ python3 -c "import torch; print(torch.__version__); print(torch.cuda.is_available())"  
2.5.0a0+872d972e41.nv24.08  
True  
jetson@yahboom:~/software/torch_pkgs$ python3 -c "import torchvision; print(torchvision.__version__)"  
0.20.0a0  
jetson@yahboom:~/software/torch_pkgs$
```

6. Check System CUDA Version

The base image for Jetson series controllers is provided by NVIDIA official, and the GPU acceleration runtime environment is uniformly managed by the JetPack package. If you need to view related version information, you can use jtop.

```
jtop
```



Click INFO to view version information

